Characterizing Fairness Trade-offs in Multi-Client Adaptive Streaming

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***Abstract*—Ensuring fairness among users competing for shared network resources remains a critical challenge in multi-client adaptive video streaming. While modern adaptation strategies optimize individual Quality of Experience (QoE), they often overlook the complex trade-offs between different fairness dimensions and playback stability. This study presents an analytical characterization of these trade-offs by evaluating six diverse adaptation strategies—ranging from heuristic-based baselines to fairness-aware reinforcement learning—under identical, highly variable network conditions. We employ a multi-metric framework that jointly analyzes four complementary fairness indicators: Time-Fair QoE (TF-QoE), QoE Fairness Score (QFS), Bitrate Fairness Index (BFI), and Stability-Aware Fairness (SAF). By analyzing experimental data from a 10-client edge-assisted testbed replaying real-world 4G/3G traces, we reveal structural conflicts between quality equity and adaptation instability. Our findings demonstrate that gains in bitrate fairness often come at the expense of increased quality switches for specific users, a trade-off that is significantly amplified under constrained network conditions. This work provides a rigorous observational baseline for the design of equitable streaming systems, highlighting that effective fairness evaluation must account for the interplay between resource distribution and temporal stability.**

***Index Terms*—Adaptive Bitrate (ABR) streaming, fairness trade-offs, edge computing, reinforcement learning, multi-metric evaluation, Quality of Experience (QoE).**

I. Introduction

The evolution of edge-assisted video delivery has transformed Adaptive Bitrate (ABR) streaming into a complex multi-agent resource contention problem. In this paradigm, the edge layer serves not only as a cache but as a critical coordination point to manage client-side conflicts. While modern adaptation logic effectively optimizes individual Quality of Experience (QoE), the structural interactions between concurrent clients sharing constrained resources remain insufficiently characterized. Traditional fairness evaluation relies heavily on aggregate indices like Jain's Fairness Index (JFI), which typically ignore the temporal and stability-related disparities that define modern edge service experience.

This study deconstructs these disparities through a multi-metric observational analysis across four dimensions: temporal quality distribution (TF-QoE), distributional fairness (QFS), bitrate allocation (BFI), and adaptation stability (SAF). By evaluating six distinct ABR strategies under real-world 4G/3G network traces, we expose the "fairness-stability trade-off"—a structural conflict where gains in resource equity often drive increased playback instability for a subset of the client fleet. Our findings indicate that network variability acts as an amplifier for these conflicts, revealing that systems appearing "fair" under stable conditions frequently exhibit severe temporal unfairness during transient congestion windows at the edge node.

The main contributions of this work are: (i) an analytical characterization of fairness-stability trade-offs explicitly tailored for edge-assisted environments; (ii) the identification of the "tactical panic" and "recovery lag" phenomena as drivers of temporal unfairness in decentralized ABR; and (iii) a set of design recommendations that demonstrate how findings from observational studies can inform the development of superior fairness-aware controllers, such as those employing edge-coordinated Reinforcement Learning (RL).

The remainder of the paper is organized as follows. Section II provides the conceptual grounding. Section III reviews the literature. Section IV describes the evaluation methodology. Section IV-D presents the analytical results. Finally, Section V concludes the paper.

II. Background

Adaptive Bitrate (ABR) streaming allows clients to dynamically select video representations based on network conditions and playback states. DASH-based implementations, such as DASH.js, typically employ heuristic rules—based on throughput estimation or buffer occupancy—to balance quality, rebuffering, and stability [1]–[3]. While effective for independent users, these strategies lead to resource contention and unfair quality allocation in multi-client environments where bandwidth is shared. Data-driven approaches, particularly reinforcement learning (RL) [4], have significantly improved individual QoE but often treat fairness as a secondary, post-hoc concern. Standard indicators like Jain's Fairness Index (JFI) [5] provide a limited view of equity, as they mask the temporal dynamics and recovery delays that define real-world user experience. Since fairness is inherently multi-dimensional, recent studies emphasize the necessity of complementary metrics that capture distributional equity, throughput allocation, and playback stability [6]–[8], motivating the joint evaluation of fairness and stability trade-offs in edge-assisted systems.

III. Related Works

While prior research established robust foundations for QoE optimization [3], [4], the interaction between multi-metric fairness and stability in multi-client scenarios remains insufficiently characterized. Previous works explored QoE fairness evaluation in edge layers [9], [10], providing the experimental basis for this study. Heuristic strategies like FESTIVE [1] and BOLA [3] improve stability but often lead to unfair resource distribution in multi-user scenarios. RL-based models like Pensieve [4] and its extensions, such as Oboe [11] and HotDASH [12], prioritize individual QoE over equity.

More recent fairness-aware metrics like QFS [7] and BFI [13] move beyond JFI but are rarely integrated into a structural analysis of adaptation trade-offs. Edge-assisted solutions like ARARAT [14] and MEC-based policies [15] investigate collaborative load balancing but still rely on aggregate indicators that mask temporal conflicts. This study deconstructs these interactions by systematically analyzing how disparate fairness metrics conflict with playback stability under identical network variability, providing the necessary evidence to design next-generation, stability-aware fairness controllers.

IV. Trade-off Analysis Methodology

The experimental data analyzed in this study are derived from controlled executions previously reported in related works [9], [10] and are reused exclusively for post-execution analysis. The analysis builds upon a validated experimental framework previously introduced by the authors [9], [10], enabling a focused investigation of fairness trade-offs without confounding architectural variations. By maintaining consistent architectural assumptions across experiments, the analysis isolates the impact of adaptation behavior and network variability on fairness–stability trade-offs in multi-client environments.

A. Experimental Scenario

We focused on multi-metric fairness trade-offs and interaction effects, without modifying the underlying adaptation mechanisms. Table II summarizes the main characteristics of the experimental scenario to ensure clarity and self-containment.

B. Overview of the Analytical Approach

The objective of the proposed methodology is to analyze trade-offs between multiple fairness metrics and Quality of Experience (QoE) indicators in multi-client adaptive streaming systems. Rather than optimizing a specific objective, the approach focuses on interpreting how different adaptation strategies influence fairness dimensions under shared network conditions. All analyses are performed offline using execution logs from controlled experimental scenarios, ensuring that adaptation decisions remain unaffected by the evaluation process.

A structured analytical workflow is adopted in which executions are grouped into observation sets, complementary fairness metrics are computed independently, inter-metric correlations are evaluated, and adaptation strategies are mapped into a fairness–stability interaction space. This enables systematic identification of metric conflicts and trade-off regions under identical network conditions.

C. Observation Sets and Experimental Grouping

To ensure consistency and interpretability, experimental results are organized into observation sets that group executions sharing comparable network characteristics. Each observation set corresponds to a specific class of network behavior, such as relatively stable conditions or highly variable throughput patterns. By avoiding the aggregation of heterogeneous scenarios, the analysis prevents misleading conclusions caused by uncontrolled variability. Within each observation set, all ABR strategies are evaluated using the same set of clients, network traces, and execution parameters. This grouping enables direct comparison of fairness outcomes and QoE stability across adaptation strategies, highlighting how identical conditions can lead to different fairness profiles.

D. Multi-Metric Fairness Characterization

To move beyond single-index evaluation, we define four fairness indicators that quantify distinct aspects of resource distribution and playback stability:

- **TF-QoE (Time-Fair QoE):** Measures the relative deviation of client i's composite QoE score (Q_i) from the session average ($\bar{Q}$):

$$TF\text{-}QoE_i = 1 - \frac{|Q_i - \bar{Q}|}{\bar{Q} + \epsilon} \tag{1}$$

  where ϵ is a small constant (10^{-6}) for numerical stability.
- **QFS (QoE Fairness Score):** An application of Jain's Fairness Index (JFI) over the temporal average of composite QoE score samples ($\hat{Q}_i$):

$$QFS = \frac{\left(\sum_{i=1}^{N} \hat{Q}_i\right)^2}{N \cdot \sum_{i=1}^{N} \hat{Q}_i^2} \tag{2}$$

- **BFI (Bitrate Fairness Index):** Evaluates fairness in quality level allocation using JFI on average numeric representation indices ($\bar{L}_i$):

$$BFI = \frac{\left(\sum_{i=1}^{N} \bar{L}_i\right)^2}{N \cdot \sum_{i=1}^{N} \bar{L}_i^2} \tag{3}$$

  where $\bar{L}_i$ ranges from 1 (360p) to 5 (2160p).
- **SAF (Stability-Aware Fairness):** Characterizes the equity of adaptation stability by considering the count of quality switches (S_i):

$$SAF = 1 - \frac{|S_i - \bar{S}|}{\bar{S} + \epsilon} \tag{4}$$

TABLE I: Comparison of Related Works from a Fairness Evaluation Perspective

Work	Adaptation Strategy	Fairness Perspective	Evaluation Limitations	Evaluation Scope
[1]	Heuristic ABR (FESTIVE)	Implicit fairness via bandwidth sharing	No explicit fairness metrics; unstable under high heterogeneity	Early analysis of fairness side-effects in ABR
[3]	Heuristic ABR (BOLA)	QoE stability-focused	Fairness not evaluated; single-client oriented	Robust buffer-based bitrate adaptation
[4]	RL-based ABR (Pensieve)	QoE-centric optimization	Fairness not considered during decision-making	Learned bitrate adaptation for individual QoE
[11]	RL-based ABR (Oboe)	Efficiency and stability	No fairness evaluation	Automated tuning of ABR parameters
[12]	Hybrid ABR (HotDASH)	Throughput efficiency	Ignores inter-client fairness	Content-aware bitrate adaptation
[7]	Fairness metric proposal	QoE distribution fairness (QFS)	Metric-level evaluation only; no ABR integration	Definition and validation of QoE fairness metric
[13]	Fairness metric proposal	Bitrate allocation fairness (BFI)	Single-metric fairness evaluation	Bitrate-based fairness assessment
[16]	Distributed bandwidth allocation	QoE-aware fairness	Focused on bandwidth allocation; limited metric diversity	Fairness-driven rate allocation at bottlenecks
[17]	Multi-agent RL-based ABR	Fairness-aware coordination	Limited fairness metrics; scenario-specific evaluation	Coordinated bitrate decisions in multi-user scenarios
[14]	Edge-assisted streaming	QoE improvement at the edge	Fairness treated implicitly or post-hoc	Edge-cloud collaboration for streaming
[15]	MEC-based optimization	QoE fairness	Analytical modeling; limited experimental analysis	Fairness-aware resource allocation in MEC
[18]	Edge-assisted ABR	Fairness via JFI	Single fairness metric; no trade-off analysis	Fairness-constrained ABR at the edge
[19]	Edge-centric heuristic	Throughput-aware fairness	Focus on estimation accuracy; ignores SAF	Edge-based quality adaptation logic
[20]	RL-based ABR (Q2D3)	QoE fairness vs preference	Scenario-specific; lacks stability trade-off	Cooperative edge server optimization
This Work	**Analytical Study**	**Multi-metric characterization**	**Observational perspective only**	**Characterizing fairness and stability trade-offs**

TABLE II: Summary of Experimental evaluation scenario.

Aspect	Description
Scenario Type	Multi-client adaptive streaming under shared edge
Clients	10 concurrent DASH clients in Docker containers
Streaming Engine	DASH.js player automated via Selenium
Network Traces	Verizon-LTE ($\mu \approx 20$ Mbps) and UMTS ($\mu \approx 2$ Mbps)
Video Content	Big Buck Bunny (6 levels: 360p–2160p)
Session	120 seconds of playback per client

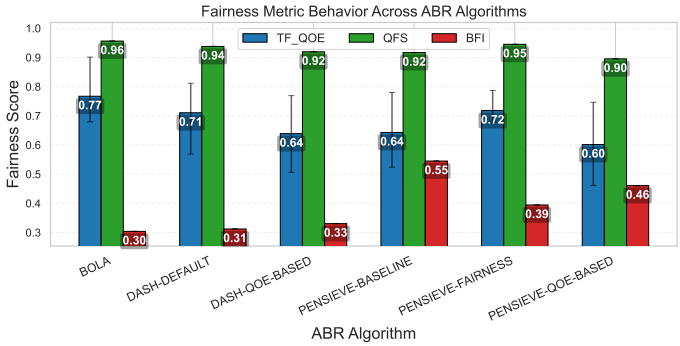

Fig. 1: Fairness metric behavior across adaptive bitrate (ABR) strategies under multi-client conditions, comparing QFS, TF-QoE, and BFI.

This section presents the results of the multi-metric fairness analysis derived from the methodology described in Section IV. The discussion moves from metric-level observations to a deeper characterization of the structural conflicts between fairness and stability, highlighting behavioral patterns that emerge under variable network conditions.

E. Fairness Metric Behavior Across Adaptation Strategies

The analysis reveals that different ABR strategies exhibit distinct fairness profiles depending on the metric considered. Metrics capturing distributional aspects of user experience, such as QoE Fairness Score (QFS), often favor strategies that prioritize stable QoE exposure, whereas metrics related to resource allocation, such as Bitrate Fairness Index (BFI), may highlight different adaptation behaviors. This divergence indicates that fairness improvements observed under one metric may coexist with degradation according to another, reinforcing the need for a multi-metric evaluation perspective.

Figure 1 illustrates this behavior under multi-client conditions. While most ABR strategies achieve consistently high QoE Fairness Score (QFS) values (typically above ≈ 0.9), substantial disparities emerge when temporal fairness (TF-QoE) and bitrate distribution fairness (BFI) are considered. In particular, TF-QoE exhibits noticeable variation across adaptation strategies, whereas BFI shows pronounced degradation for algorithms that prioritize individual QoE maximization, with values dropping to the 0.7–0.8 range in some cases. These results indicate that strategies maintaining stable QoE exposure across users do not necessarily ensure equitable bitrate allocation, especially under heterogeneous network conditions. Consequently, single-metric fairness evaluation may obscure relevant trade-offs, reinforcing the need for a multi-dimensional fairness analysis in adaptive streaming systems.

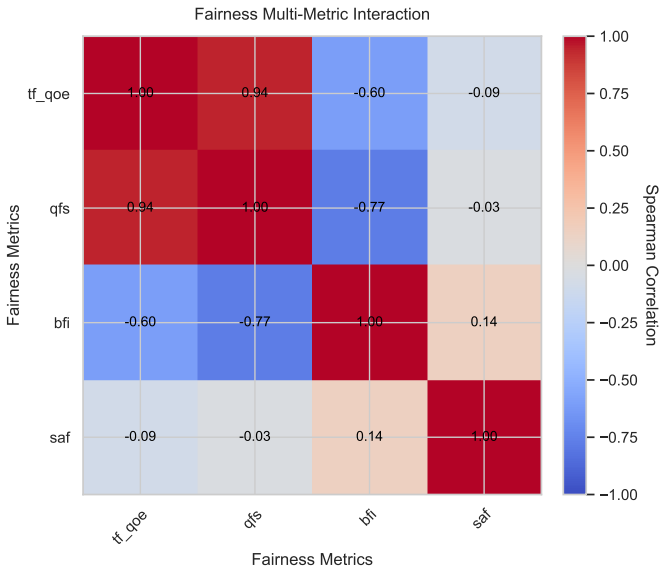

Fig. 2: Multi-metric fairness interaction under multi-client conditions, based on Spearman correlation analysis.

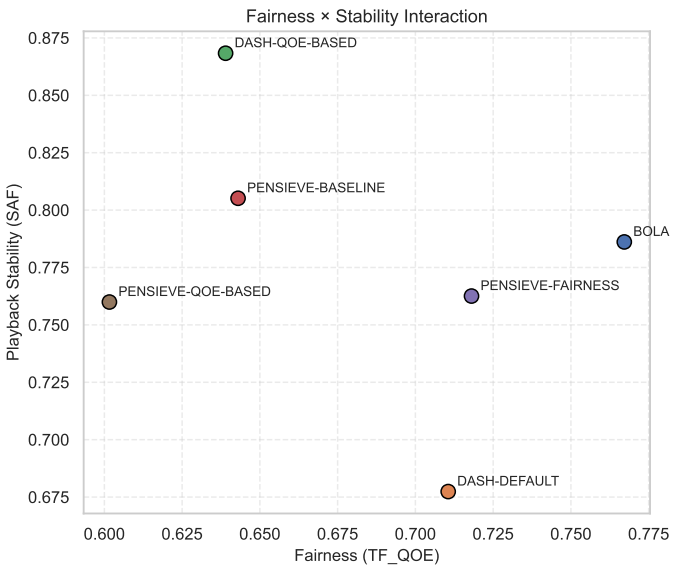

Fig. 3: Fairness versus playback stability interaction under multi-client conditions, relating TF-QoE and SAF across ABR strategies.

F. Multi-Metric Fairness Interaction

Figure 2 analyzes the interaction between different fairness metrics under multi-client conditions using Spearman correlation analysis. The results reveal a strong positive correlation between TF-QoE and QoE Fairness Score (QFS) ($\rho \approx 0.94$), indicating that both metrics capture closely aligned aspects of perceived user experience fairness. In contrast, pronounced negative correlations are observed between QoE-oriented fairness metrics and bitrate distribution fairness. Specifically, QFS exhibits a strong negative correlation with the Bitrate Fairness Index (BFI) ($\rho \approx -0.77$), while TF-QoE shows a moderate negative correlation with BFI ($\rho = -0.60$). These results indicate that improvements in perceived QoE fairness are often accompanied by increased imbalance in bitrate allocation among competing clients. Stability-Aware Fairness (SAF), in turn, presents only weak correlations with the remaining metrics ($|\rho| \leq 0.14$), suggesting that adaptation stability captures a largely orthogonal fairness dimension. Overall, these interactions demonstrate that fairness metrics do not evolve independently and may encode competing objectives, particularly in heterogeneous and congested environments. Consequently, aggregated or single-metric fairness evaluation may obscure underlying conflicts between fairness dimensions, reinforcing the need for multi-metric analysis when interpreting adaptive streaming behavior.

G. Trade-offs Between Fairness and QoE Stability

A recurring pattern observed across scenarios is the interaction between fairness outcomes and QoE stability indicators, such as quality oscillations and rebuffering behavior. Figure 3 illustrates this interaction by jointly analyzing temporal fairness (TF-QoE) and playback stability (SAF) under multi-client conditions. The results show that adaptation strategies occupy distinct regions of the fairness–stability space, indicating that improvements along one dimension do not necessarily translate into gains along the other. Adaptation strategies that aggressively react to short-term throughput variations may improve fairness-related metrics but often at the cost of increased playback instability. As illustrated in Figure 3, DASH-DEFAULT achieves moderate temporal fairness (TF-QoE ≈ 0.71) but exhibits the lowest playback stability among the evaluated strategies (SAF ≈ 0.68). In contrast, DASH-QoE-BASED prioritizes stability, reaching the highest SAF value (SAF ≈ 0.87), while operating at lower temporal fairness levels (TF-QoE ≈ 0.64). RL-based strategies occupy intermediate regions of the fairness–stability space. For instance, PENSIEVE-FAIRNESS improves temporal fairness relative to the baseline (TF-QoE ≈ 0.72) while maintaining moderate stability (SAF ≈ 0.76), whereas BOLA simultaneously achieves high fairness and stability (TF-QoE ≈ 0.77, SAF ≈ 0.79). These results confirm that fairness and stability objectives are not always aligned and that their trade-offs become more pronounced under heterogeneous and variable network conditions.

H. Impact of Network Variability on Fairness Trade-offs

Network variability acts as a catalyst for fairness degradation, specifically by triggering asymmetric adaptation cycles. Figure 4 illustrates the behavior under the Verizon-LTE trace. During throughput dips (e.g., the 40s mark), reactive strategies exhibit a ”tactical panic” where standard deviation spikes exceed 10 Mbps. This indicates that while the total bandwidth is equally reduced, the *rate* of quality downswitching varies significantly across clients due to slightly different buffer pres-

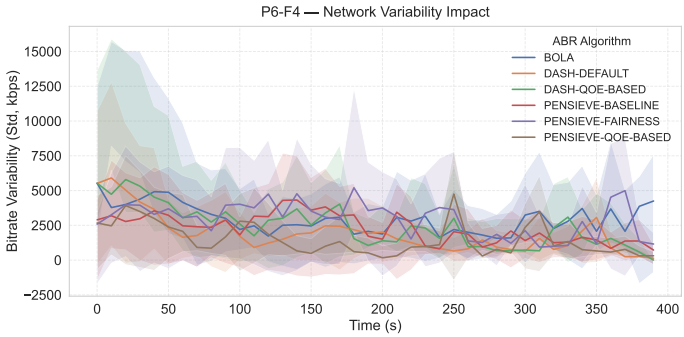

Fig. 4: Network variability impact on bitrate adaptation over time under multi-client conditions. The figure shows standard deviation of bitrate selection across 10 clients replaying the Verizon-LTE trace.

sures. Furthermore, we identify a "recovery lag" phenomenon visible after the 80s mark: as bandwidth stabilizes, individual client analysis (e.g., Client 1 vs Client 8) reveals that while Client 1 aggressively recovers to 2160p within 5 seconds, Client 8 remains "locked" at 360p/720p for up to 20 additional seconds to replenish its buffer. This behavioral heterogeneity creates a persistent gap in TF-QoE that distributional metrics like QFS or JFI completely obscure, proving that recovery lag is a critical structural unfairness in decentralized ABR logic.

I. Implications for Fairness-Aware ABR Design

Our results suggest that achieving holistic fairness requires moving beyond individual bitrate maximization. We propose three actionable design shifts:

- **Inter-Client Stability Constraints**: ABR controllers should incorporate the variance of switch counts $\sigma(S)$ into their objective functions. Designers should penalize strategies where specific users suffer disproportionate quality oscillations compared to the fleet average (SAF-aware control).
- **Synchronized Recovery Windows**: To mitigate the TF-QoE "recovery lag," edge-assisted controllers should utilize cross-client synchronization to coordinate quality upswitches, preventing a subset of clients from monopolizing throughput during bandwidth recovery.
- **Asymmetric Reward Penalties**: In cellular-dominant environments, fairness-aware RL reward functions (R) should prioritize the standard deviation of perceived QoE over the session average ($\bar{Q}$), explicitly treating fairness as a stability-linked temporal dimension.

J. Discussion and Implications

From an analytical perspective, the results show that fairness trade-offs are structural properties emerging from the interaction between adaptation logic and network variability, rather than anomalies of specific algorithms. Fairness in multi-client adaptive streaming is inherently multi-dimensional, and no single ABR strategy consistently dominates across all fairness metrics and QoE stability indicators. Instead, strategies occupy distinct regions of the fairness–stability space, reflecting implicit trade-offs between responsiveness and conservatism. Improvements in distributional fairness, bitrate allocation, or temporal QoE exposure do not necessarily occur simultaneously. Aggressive adaptation may enhance certain fairness metrics while degrading playback stability, whereas conservative strategies preserve stability at the cost of increased long-term disparities. Network variability amplifies these effects: under stable conditions, fairness outcomes tend to converge, while highly variable environments expose latent conflicts between fairness dimensions. These findings highlight the limitations of single-metric evaluation and carry direct implications for fairness-aware ABR design. Single-metric optimization is structurally insufficient in multi-client environments, and adaptation mechanisms—particularly reinforcement learning policies—should incorporate multi-dimensional evaluation criteria that explicitly account for network variability as a fairness amplifier.

A critical aspect of our evaluation is the balance between scale and architectural fidelity. While simulators (e.g., ns-3 or Python-based ABR simulators) can handle hundreds of agents, they often abstract away the system-level overheads of real video players. Our choice of 10 concurrent clients—each running a full Chromium instance via Selenium and a high-fidelity DASH.js player—ensures that our findings capture real-world resource contention (CPU/Memory cycles) that is absent in simulation. These implementation-heavy dynamics are essential for characterizing the "tactical panic" phenomena described in Section IV-D. For larger scales (50+ users), our analytical framework suggests that edge-assisted coordination becomes mandatory to mitigate the resource-heavy penalties of decentralized adaptation, providing a clear roadmap for future 5G-Advanced and 6G streaming optimization [21].

V. Conclusion and Future Work

This study demonstrates that fairness trade-offs are structural properties of multi-client adaptive streaming systems, rather than anomalies of specific algorithms. Through a multi-metric observational analysis, we show that fairness is inherently multi-dimensional and cannot be adequately captured by single or aggregated indicators. Different ABR strategies occupy distinct regions of the fairness–stability space, reflecting structural trade-offs between responsiveness and conservatism under shared conditions. Our results reveal that network variability acts as a fairness amplifier, exposing latent conflicts that remain obscured under stable scenarios. Rather than identifying a universally optimal strategy, this work emphasizes the need for structured, multi-dimensional evaluation frameworks that account for interacting metrics and network dynamics. Future work will investigate how these analytical insights can inform the design of edge-assisted, multi-objective learning strategies and coordinate recovery windows to balance competing fairness dimensions in next-generation 5G-Advanced and 6G environments.

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